PwC UK Office Attendance Predictor

Project Proposal

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1. Problem Definition

PwC UK operates a hybrid working model, so daily office attendance varies considerably by site and weekday. Facilities Management (FM) uses attendance to plan catering and cleaning, and Real Estate uses it to allocate space across teams. With little forward visibility, FM risks food waste, mis-sized cleaning shifts and shortfalls on busy days.

Attendance is also shifted by external events such as bank holidays and rail strikes, which are currently logged through a manual form that is slow and misses events.

The work sits within the Real Estate Data Analytics (REDA) team, whose Databricks lakehouse holds aggregated daily headcount per office. Forecasting is currently limited to Power BI's built-in forecast, which has not been validated. Without a validated forecast, FM cannot quantify the benefit of earlier planning, so any model must beat the heuristic of assuming attendance equals the same weekday last week.

2. Project Aim and Objectives

Aim: To develop and evaluate a 1-14 day office attendance forecasting pipeline that outperforms seasonal naive and Power BI baselines, ingests public disruption data automatically and provides calibrated prediction intervals.

Objectives:

1. Agree KPIs, horizon and simulation assumptions.
2. Automate ingestion of bank holidays, strikes and weather warnings into the existing medallion architecture.
3. Benchmark seasonal naive and Power BI forecasts as baselines.
4. Compare SARIMAX, LightGBM and XGBoost using rolling-origin validation per office, measured by MAE and MAPE.
5. Identify which events most shift attendance using feature importance.
6. Deliver a dashboard for FM and Real Estate stakeholders and estimate the potential planning benefit.
7. Evaluate results and recommend next steps.

Future value (out of scope): If validated on real data, a sufficiently accurate forecast could support exploring the subletting of floors in larger offices on low attendance days, similar to flexible workspace providers such as WeWork.

3. Research Plan

Prior work will be identified through Google Scholar, IEEE Xplore, the ACM Digital Library, ScienceDirect and arXiv, accessed via the QMUL library, plus Databricks and MLflow documentation for implementation practice. Indicative search terms include "office occupancy forecasting", "hybrid work space utilisation", "time series forecasting exogenous events", "gradient boosting forecasting", "rolling origin evaluation", "conformal prediction time series" and "forecast accuracy metrics".

Peer-reviewed work from 2018 onwards will be prioritised for methods, and post-2020 studies for hybrid working behaviour. Grey literature such as consultancy reports will be labelled and treated cautiously, with quality judged on peer review, data transparency and relevance to office settings. Findings will inform model selection, feature design, validation strategy and interval calibration.

AI tools may support discovery and summarisation, but every source will be read and verified directly and managed in mybib, with no AI-generated references.

4. Identification of Technologies, Tools and/or Datasets

Platform: Databricks with Delta tables and MLflow, chosen because the data already resides there and Unity Catalog provides governed access, lineage and experiment tracking. Alternatives are local Python with DuckDB (cheaper, but ungoverned) and Azure ML (heavier infrastructure for a small problem).

Models: Python with statsmodels for SARIMAX (an interpretable classical benchmark) (Hyndman and Athanasopoulos, 2021), plus LightGBM (Ke et al., 2017) and XGBoost (Chen and Guestrin, 2016), which handle exogenous event features flexibly. Prophet is excluded to limit scope, and deep learning because daily office series are small and interpretability matters to FM.

Data: Synthetic data is the primary dataset, generated in Python with Faker and custom simulation code. It mirrors the schema of the REDA attendance, office and date tables, with attendance patterns and event effects simulated from documented assumptions. Bank holidays, Met Office warnings, TfL disruption and strike dates are real public data. No real PwC attendance data is used.

5. Project Plan and Milestones

5.1. Key dates

Status	Milestone	Date
Met	Project options shortlisted	Oct 2026
Planned	Discuss options with manager and lecturer	TBC
Planned	Proposal submitted (IOT6002U 001)	30 Oct 2026
Planned	AI pitch video (IOT6000U)	5 Nov 2026
Planned	Project build and evaluation (16 weeks, assumed)	Start TBC
Planned	Gateway, then project report (6000 words), presentation and KSB mapping	TBC

5.2. Project build schedule (weeks from project start, assumed 16)

Task	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
KPIs, horizon and simulation assumptions	■	■														
Synthetic data generator		■	■	■												
Ingestion pipeline (bronze to gold)			■	■	■	■										
Baselines (seasonal naive, Power BI)					■	■										
Model benchmarking in MLflow							■	■	■	■						
Intervals and feature importance											■	■				
Dashboard and benefit estimate													■	■		
Evaluation and recommendations															■	■

Dependencies: modelling depends on the synthetic data generator and ingestion pipeline. The dashboard depends on the chosen model and validated intervals. The report is written after the gateway, outside this schedule.

6. Risk Register

ID	Risk	Likelihood	Severity	Mitigation
R1	Synthetic data too simple, so models only recover the generating rules	High	High	Add noise, regime changes and unmodelled events, hold out event types from training, document all generator assumptions in an appendix, state that results show method performance rather than real-world accuracy
R2	Real PwC values or identifiers included by accident	Low	High	Generate from schema only, copy no real values, review all outputs before sharing
R3	Findings do not transfer to real data	Medium	Medium	Frame as a method evaluation, build the pipeline so real data can be swapped in later, base parameters on qualitative patterns described by FM
R4	Public feeds unreliable or change format	Medium	Medium	Store raw responses in bronze, validate on load, fall back to manual flag
R5	Models fail to beat baselines	Medium	Medium	Report as a valid finding, recommend baseline-plus improvements
R6	Scope creep	Medium	Medium	Fixed objectives, MCP and Genie work out of scope
R7	Confidentiality limits on describing PwC systems	Medium	Medium	Agree restrictions with manager on Power BI setup, office counts and schemas, use generic names
R8	Power BI baseline run on synthetic data is only a proxy for the live forecast	Medium	Low	Document the settings used, state the limitation in the evaluation
R9	Compute cost overrun	Low	Low	Small data, job clusters with auto-termination

Ethics and compliance: The project uses synthetic data generated from table schemas only, with no real PwC values and no human participants. QMUL ethics requirements will be confirmed with the module organiser, and confidentiality limits on describing PwC systems will be agreed with the line manager.

7. References

- Chen, T. and Guestrin, C. (2016) 'XGBoost: a scalable tree boosting system', in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. New York: ACM, pp. 785-794.
- Hyndman, R.J. and Athanasopoulos, G. (2021) *Forecasting: principles and practice*. 3rd edn. Melbourne: OTexts.
- Ke, G. et al. (2017) 'LightGBM: a highly efficient gradient boosting decision tree', in *Advances in Neural Information Processing Systems 30*. Red Hook, NY: Curran Associates, pp. 3146-3154.

8. Appendices

8.1. Appendix A: Generative AI declaration

Generative AI was used in order to help explore potential risks, as well as non-trivial formatting (typst layout). All AI output was read and verified before being inputted into my project proposal. I take full responsibility for the final content of this document.

8.2. Appendix B: Planned KSB mapping

KSB	**Planned evidence**
S50	FM dashboard and communication of results
S52, S53	Model benchmarking, rolling-origin validation and stability testing
S54, K56	Automated ingestion of public data sources
S55, K54	Analysis of simulated attendance across offices
S2, K3	Risk register and mitigations
S5, S6, K15	Project plan, schedule and time/cost estimation
K18, S14	Research plan and evaluation of forecasting methods
B3	Synthetic data approach, no real PwC data used